

Product Requirements Document (PRD)

Project: UniThrift | The Ultimate Campus Ecosystem (GTBIT / GGSIPU)

1. Executive Summary

UniThrift is a hyper-local, exclusive campus ecosystem designed for university students. Expanding beyond a simple peer-to-peer marketplace for academic gear, UniThrift integrates B2B PG/Hostel listings and a community-driven Flatmate Finder. Access is strictly gatekept via GGSIPU enrollment numbers for students, ensuring absolute trust. Monetization scales from micro-transactions on student gear to premium, recurring subscription fees from local PG owners.

2. Core Pillars & Features

2.1 Trust & Verification (The Gatekeeper)

- **Student Onboarding:** Users register via their 11-digit GGSIPU enrollment number. The system auto-extracts admission year and branch to create uneditable trust badges (e.g., "1st Year - CSE"). Manual student ID photo upload is required for final verification.
- **Business Onboarding:** PG owners register with a phone number and business details (no enrollment number required). They receive a distinct "Verified PG" badge after manual admin vetting.

2.2 Pillar 1: The Thrift Marketplace

- **Focus:** Books, Drafters, Calculators, Lab Coats.
- **Monetization (WhatsApp Paywall):** No in-app chat. Buyers pay ₹50 via Razorpay to "Reserve" an item for 48 hours. This payment reveals a wa.me link to directly WhatsApp the seller.
- **Revenue Split:** ₹35 goes to the seller (manual disbursement), ₹15 retained by the platform.

2.3 Pillar 2: PG & Hostel Hub (B2B Revenue)

- **Focus:** Verified local housing options targeted directly at first and second-year students.
- **Features:** Image carousels, rent pricing, amenity icons (WiFi, AC, Meals), and distance from GTBIT campus.
- **Monetization:** "Sponsored Listings." PG owners pay ₹999/month to pin their properties to the top of the feed during peak admission season.

2.4 Pillar 3: Flatmate Finder (Community Growth)

- **Focus:** A safe, enrollment-verified matching system for students seeking roommates.
- **Features:** Profile cards displaying max budget, preferred locations (e.g., Janakpuri, Dwarka), and lifestyle tags (e.g., "Night Owl", "Non-Smoker").
- **Goal:** High-retention growth hack to keep students opening the app daily.

3. Tech Stack

Layer	Technology	Rationale
Frontend	Next.js + Tailwind CSS	Mobile-first, SEO-friendly, fast rendering.
Backend / Auth	Supabase (PostgreSQL)	Handles complex user roles (Student vs PG) and RLS security.
Payments	Razorpay API	UPI intent links for ₹50 micro-transactions & B2B subscriptions.
Automation	Python (FastAPI)	OCR for Student ID verification and admin dashboards.

4. Database Architecture (Supabase)

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-- 1. USERS (Handles both Students and Businesses)
id (UUID, PK)
role (String) -- 'student', 'pg_owner', 'admin'
full_name (String)
phone_number (String)
enrollment_number (String, Unique, Nullable)
branch_code (String, Nullable)
admission_year (Int, Nullable)
is_verified (Boolean)

-- 2. THRIFT LISTINGS (Pillar 1)
id (UUID, PK)
seller_id (UUID, FK -> Users)
title (String) | price (Decimal) | category (String)
status (String) -- 'active', 'on_hold', 'sold'

-- 3. PG LISTINGS (Pillar 2)
id (UUID, PK)
owner_id (UUID, FK -> Users)
pg_name (String) | description (Text) | price_per_month (Decimal)
amenities (Array) | distance_from_college (Decimal)
is_sponsored (Boolean)

-- 4. FLATMATE PROFILES (Pillar 3)
id (UUID, PK)
user_id (UUID, FK -> Users)
budget_max (Decimal)
preferred_locations (Array) | lifestyle_tags (Array)
status (String) -- 'looking', 'found'
```